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MAML-Integrated multi-agent reinforcement learning for adaptive coalition-based UAV coordination in disaster scenarios

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ABSTRACT

Unmanned aerial vehicles (UAVs) are increasingly used for time-critical disaster response tasks such as search, rescue, and situational assessment. However, coordinating heterogeneous UAV swarms remains challenging due to dynamic task demands, limited onboard resources, intermittent communication, noisy sensing, and platform failures. This paper presents a resource-aware coalition-based task assignment and path-planning (RCTP) framework that integrates model-agnostic meta-learning (MAML) with multi-agent deep deterministic policy gradient (MA-DDPG) to enable rapid adaptation and scalable decentralized coordination in realistic disaster environments. RCTP introduces an end-to-end learning-based coordination framework for multi-UAV disaster response, where MAML provides meta-initialized policies that adapt to new disaster scenarios with only a few gradient updates, while MA-DDPG supports continuous-action control for coalition formation, task assignment, and energy-aware path planning. The framework explicitly models UAV heterogeneity, intermittent LoS/NLoS communication, dynamic obstacles, noisy observations, and UAV failures. Coalition formation is guided by interpretable and resource-aware suitability scores that are updated online, enabling decentralized task reassignment, fault tolerance, and load balancing without relying on centralized infrastructure. Extensive simulations with 10–30 UAVs across heterogeneous tasks show that RCTP outperforms the existing schemes (i.e., PPO, DQN, and MA-DDPG), achieving 30–40% faster mission completion and 10–20% lower energy consumption, higher task success rates, and improved robustness under multiple failures. Complexity analysis confirms lightweight online execution with millisecond-level per-agent inference, demonstrating scalability and real-time feasibility for disaster response. These results establish RCTP as a practical, robust, and adaptive solution for multi-UAV disaster response.

Introduction

Unmanned aerial vehicles (UAVs) are increasingly deployed in time-critical disaster response missions, including search and rescue, damage assessment, and emergency delivery, where rapid coverage, access to hazardous zones, and real-time sensing are essential [1]. Coordinated UAV swarms further increase these advantages by enabling parallel task execution, improving spatial

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coverage, and maintaining mission continuity even under individual UAV failures [2]. Despite this potential, reliable multi-UAV coordination in real disaster environments remains challenging. Practical deployments must operate under limited onboard energy, intermittent and non-line-of-sight (LoS/NLoS) communications, dynamic obstacles and no-fly regions, noisy sensing, and stochastic UAV failures. At the same time, they must support timely prioritization of heterogeneous and evolving tasks in real-time [3]. These constraints make robust, scalable, and adaptive coordination particularly difficult in realistic disaster scenarios. Existing multi-UAV coordination approaches, including rule-based heuristics, optimization-driven methods, and centralized learning-based frameworks, often address isolated components such as task assignment, path planning, or energy optimization. Many of these methods require scenario-specific tuning, rely on centralized controllers or additional infrastructure, or assume fixed task distributions and stable communication conditions [4,5]. Although recent multi-agent reinforcement learning (MARL) approaches improve adaptability and reactivity, they typically lack mechanisms for fast adaptation to new environments, fault-tolerant coalition restructuring, and scalable real-time execution under heterogeneous resources and intermittent connectivity. As a result, their effectiveness in realistic disaster settings remains limited.

To address these limitations, this paper proposes a resource-aware, coalition-based task assignment and path-planning (RCTP) framework for robust multi-UAV coordination in disaster environments [6,7]. RCTP provides an end-to-end learning-based coordination framework that jointly integrates coalition formation, task allocation, and path planning within a unified framework, rather than optimizing a single coordination subproblem in isolation. The framework combines model-agnostic meta-learning (MAML) with multi-agent deep deterministic policy gradient (MA-DDPG) to achieve rapid adaptation and decentralized continuous control without reliance on centralized infrastructure. During offline training, MAML learns a meta-initialized policy across diverse disaster scenarios, enabling each UAV to adapt quickly to new task distributions and environmental conditions using only a few gradient updates. During online execution, MA-DDPG supports stable continuous-action control for coalition formation, priority-aware task assignment, and energy-efficient path planning under safety and connectivity constraints [8].

Coalitions in RCTP are formed using an interpretable, resource-aware suitability score that jointly accounts for UAV heterogeneity, including remaining energy, sensing payloads, proximity to tasks, communication quality, and task requirements [9,10]. A priority-aware reward structure promotes urgent task execution while maintaining balanced workloads across agents [11]. The framework further incorporates UAV health monitoring, failure detection, dynamic task reassignment, and connectivity-aware replanning, allowing missions to continue seamlessly under UAV failures, link degradation, and dynamic obstacles [12–16]. Importantly, RCTP explicitly models realistic disaster conditions, including static and dynamic obstacles, LoS/NLoS communication, noisy observations, and heterogeneous UAV capabilities. Extensive simulations are conducted under varying swarm sizes, task distributions, and environmental conditions. The results demonstrate consistent improvements over strong coordination baselines in task completion time, energy efficiency, load balancing, and robustness to UAV failures. Moreover, RCTP exhibits rapid adaptation to previously unseen scenarios via few-shot updates, supporting fast deployment in new disaster environments [17,18]. Overall, these results

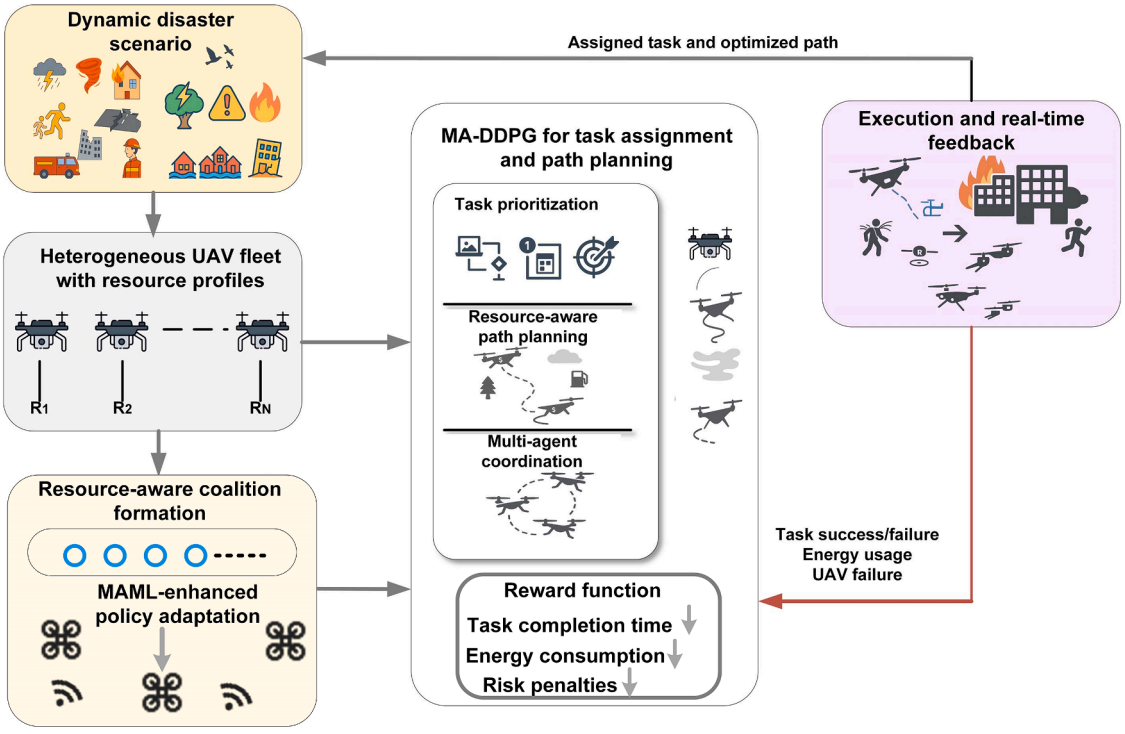


Fig. 1. Coalition-based task assignment and path planning architecture for multiple UAVs in the RCTP framework.

indicate that RCTP provides a practical, scalable, and adaptive solution for resilient multi-UAV coordination in complex real-world disaster settings. An overview of the RCTP system model and its key operational components is shown in Fig. 1.

Contributions of the paper

This paper presents RCTP, a learning-based framework for resilient and adaptive coordination of heterogeneous UAV swarms in disaster response. The main contributions are summarized as follows:

- **Meta-adaptive multi-agent coordination**: RCTP tightly integrates MAML with MA-DDPG for multi-UAV disaster response, enabling agents to rapidly adapt to new task distributions and environmental conditions with only a few gradient updates, rather than retraining from scratch as in conventional MARL methods.
- **Resource-aware coalition formation with interpretable scoring**: Unlike existing coalition or task-assignment methods that assume homogeneous agents or static capabilities, RCTP forms coalition using an interpretable suitability score that explicitly captures UAV heterogeneity (including energy state, sensing payloads, communication quality, and proximity), allowing coalitions and roles to be dynamically reconfigured as mission conditions evolve.
- **Priority-aware and energy-constrained task assignment:** RCTP introduces a joint prioritization-feasibility scheduling mechanism developed through a tailored reward formulation that balances urgent task completion with energy efficiency, safety, and coordinated coalition behavior, ensuring mission effectiveness under severe resource constraints.
- **Continuous-control path planning under realistic constraints**: By leveraging continuous-action MA-DDPG, RCTP avoids control discretization and jointly optimizes energy consumption, latency, collision avoidance, connectivity maintenance, and obstacle avoidance, addressing limitations of PPO/DQN-based approaches in complex 3D environments.
- **Online fault tolerance and dynamic replanning**: RCTP incorporates real-time UAV health monitoring, failure detection, and immediate task reassignment with connectivity-aware path replanning, enabling uninterrupted mission execution under UAV dropouts and link degradation without centralized control or external infrastructure.
- **Scalable and real-time feasible design**: The framework separates offline meta-training from lightweight online execution, achieving millisecond-level per-agent inference, which makes RCTP suitable for deployment on resource-constrained UAV platforms and scalable to larger swarms.
- **Comprehensive evaluation under realistic disaster conditions**: Extensive simulations with varying swarm sizes, heterogeneous tasks, dynamic obstacles, NLoS communication, and UAV failures demonstrate that RCTP consistently outperforms strong baselines in terms of mission completion time, energy efficiency, fairness, and robustness.

Organization of the paper

The remainder of this paper is structured as follows. Section 2 reviews the literature on UAV coordination, task allocation, and reinforcement learning in disaster response scenarios. Section 3 introduces the foundational concepts of our approach, detailing the theoretical background of MARL and the meta-learning techniques employed, along with the key principles and algorithms that enable efficient policy learning and rapid adaptation in dynamic environments. Section 4 addresses the system model and problem formulation. Section 5 presents the proposed RCTP algorithm, outlining its coalition formation, task assignment, and path planning strategies. Section 6 reports performance simulation results under diverse conditions. Finally, Section 7 concludes the paper and discusses potential avenues for future research.

Review of related literature

Research on multi-UAV systems has extensively explored coalition formation, task assignment, and path planning to improve coordination efficiency in complex missions. Early approaches primarily relied on rule-based heuristics, optimization techniques, and game-theoretic formulations to distribute tasks and resources among UAVs. While effective in structured settings, these methods often assume static environments, homogeneous platforms, or centralized control architectures, limiting their applicability to real-world disaster scenarios characterized by uncertainty, heterogeneity, and frequent disruptions. This section reviews recent research on UAV coalition formation, task assignment, and path planning, with emphasis on intelligent and learning-based approaches for dynamic and resource-constrained environments.

Coalition formation and cooperative UAV path planning

Coalition formation has been widely studied to enable cooperation in UAV swarms. Distributed coalition schemes [19–21] improve fairness and resource utilization, while game-theoretic and multi-robot coalition models [22–24] explicitly model task dependencies and competition. Many existing multi-UAV coordination and task allocation approaches assume homogeneous UAV capabilities, static coalition structures, or reliable communication, which limit their adaptability in disaster environments characterized by heterogeneous platforms, dynamic priorities, and frequent failures. Recent context-aware coverage path planning methods (including those using mobile ground stations for battery swapping) jointly optimize coverage, trajectories, and energy replenishment and are effective in structured or semi-static scenarios [25]. However, these approaches typically rely on dedicated ground infrastructure, centralized coordination, and continuous charging availability, increasing system complexity and reducing applicability in fast-evolving disaster

settings. In contrast, RCTP targets infrastructure-free operation and forms heterogeneous and resource-aware coalitions that are dynamically reconfigured based on real-time energy state, sensing payloads, communication quality, and task requirements, enabling scalable, fault-tolerant coordination under severe resource, and connectivity constraints.

Task assignment using reinforcement learning

Reinforcement learning has been increasingly applied to UAV task assignments. Deep RL methods [13,26,27] learn allocation policies from interaction data, while priority-aware strategies [10,12,28,29] emphasize urgent tasks. However, most RL-based approaches are trained under fixed task distributions and require extensive retraining when conditions change. Moreover, UAV heterogeneity and failure handling are often neglected. RCTP addresses these gaps by integrating MAML into multi-agent RL, enabling rapid adaptation to new task sets and environments with minimal additional training.

Intelligent and learning-based path planning techniques

Path planning techniques based on particle swarm optimization (PSO) [30], genetic algorithms (GA) [31], and MARL [32] are effective in static or moderately dynamic environments but often lack real-time replanning, task awareness, and resource-constrained operation. Digital-twin-based [33] and bio-inspired methods [34] improve solution quality but incur high computational overhead or require additional infrastructure. RCTP instead employs continuous-control MARL with online replanning, enabling obstacle-aware and energy-efficient trajectories tightly coupled with coalition decisions and task priorities.

Integrated task and path optimization

Several studies jointly optimize task assignment and path planning using MARL [32,35], by improving coordination in uncertain environments. While these approaches demonstrate improved learning efficiency, they typically rely on single-level training and lack meta-learning capabilities or explicit fault-tolerance mechanisms [36]. As a result, their ability to generalize to unseen disasters, communication degradation, and UAV failures remains limited. RCTP extends this line of work by embedding meta-learning within an actor-critic framework, enabling UAVs to generalize from prior experience and adapt rapidly to dynamic tasks, link variability, and platform failures.

Novelty of RCTP

Unlike recent infrastructure-dependent or computationally intensive approaches, RCTP provides an end-to-end, learning-based solution that jointly addresses coalition formation, task assignment, and path planning with rapid adaptability and real-time feasibility in dynamic disaster environments. Rather than relying on external energy replenishment or centralized coordination, RCTP emphasizes energy-aware task allocation, coalition-based load balancing, decentralized execution, and fast adaptation through meta-learning. This design enables infrastructure-free deployment, improved robustness under intermittent communication, and scalable coordination under heterogeneous resources, all of which are key requirements for real-world disaster response. Table 1 summarizes representative related works and highlights how RCTP advances adaptability, resilience, and operational efficiency compared to existing methods.

Preliminaries

This section outlines the fundamental concepts of the proposed RCTP framework. It begins with the operational constraints that affect UAV coalition behaviour and then details the challenges of MARL environments. It also introduces the integration of MAML into the MA-DDPG framework to enhance policy adaptability. A complete list of notations used throughout this paper is provided in Table 2.

Constraints in coalition formation, task assignment, and path planning

RCTP coordinates heterogeneous UAVs under realistic disaster conditions characterized by limited energy and payload capacity, intermittent LoS/NLoS communication, dynamic obstacles and no-fly regions, evolving task priorities, and partial observability with noisy state information. The system must also tolerate sudden UAV failures and maintain mission progress under resource scarcity and connectivity disruptions. These constraints require interpretable and resource-aware coalition formation, priority- and feasibility-driven task assignment, and energy-efficient, collision-free, and connectivity-aware path planning. RCTP explicitly incorporates these constraints to enable adaptive, fault-tolerant, and scalable multi-UAV coordination in dynamic disaster environments.

Challenges in multi-agent learning

Multi-UAV coordination is difficult due to partial observability, dynamic environments, heterogeneous UAV capabilities, and complex interactions among agents. RCTP addresses these challenges using MA-DDPG with decentralized actors and a centralized critic, enabling continuous control without the performance loss caused by action discretization in PPO or DQN-based methods. To

Table 1
Comparative overview of related studies.

Ref.	Algorithm Used	Main Idea	Application Domain	Key Limitations	CF	TA	PP
[24]	Interference-aware path planning	Routing under interference constraints	Beyond-5G UAV networks	Navigation only; no task or coalition modeling	No	No	Yes
[19,20]	Energy-efficient strategies	UAV endurance optimization	General UAV operations	Static task models; no learning or adaptation	Yes	Yes	No
[22]	Distributed coalition schemes	Fair resource sharing	Resource management	No task prioritization or path planning	Yes	Yes	No
[21]	Multi-robot coalition models	Cooperative task execution	Collaborative robotics	Assumes static environments; no dynamic replanning	Yes	No	No
[23]	Resource-aware matching	Capability-to-task matching	Task allocation	Poor adaptability to dynamic scenarios	No	Yes	No
[32]	Digital twin-assisted optimization	Task assignment via digital replicas	Task assignment	High computational overhead; unsuitable for real-time disaster response	No	Yes	No
[35]	Distributed timing algorithms	Deadline-aware task execution	Time-critical missions	No learning or path planning capability	No	Yes	No
[31]	MARL	Joint task allocation and routing	UAV navigation	Requires extensive retraining; limited adaptability	No	Yes	Yes
[33]	Nature-inspired optimization	Assignment and routing	UAV navigation	Heuristic-based; limited generalization	No	No	Yes
[34]	Task coordination under uncertainty	Large-scale task allocation	Multi-UAV systems	No end-to-end learning or rapid adaptation	No	Yes	Yes
[10,12]	Priority-aware heuristics	Importance-based task assignment	Multi-UAV systems	No dynamic reassignment or learning	No	Yes	Yes
[14]	Coalition formation game	Strategic coalition optimization	Electronic warfare	Static game assumptions; no real-time learning	Yes	Yes	No
[13,25–27]	RL-based (DQN, PPO, MADDPG)	Learning-based tasks and path planning	Multi-UAV systems	Slow adaptation; no meta-learning	No	Yes	Yes
[28]	Deep RL	Communication-centric optimization	UAV networks	Limited to data harvesting; no coordination	No	No	No
[29]	PSO	Metaheuristic routing	Network optimization	No task or coalition modeling	No	No	Yes
[30]	Genetic algorithm	Task sequencing	Task scheduling	Ignore coordination and navigation	No	Yes	No

Note: CF: Coalition formation; TA: Task assignment; PP: Path planning.

Table 2
List of symbols and parameters.

Notation	Description	Notation	Description
U, T, A	Sets of UAVs, tasks, task areas	$\mathrm{p}_i(t)$	Position of UAV i at time t
N, M	No. of UAVs, tasks	D_{ij}, d_{min}	Distance, minimum separation
R, L_i	Communication range, load of UAV i	ψ[t], θ[t]	Heading, elevation angles
$B(u_i)$	Battery (energy) of UAV i	$R_{rem(i)}$	Remaining resource of UAV i
R_j^{req}	Resource need of task j	$cap(i)$	Capacity of UAV i
$\boldsymbol{X_{ij}} \in \{0,1\}$	Assignment indicator $(i \rightarrow j)$	S_{ij}, e_{ij}	Suitability score, displacement estimation error
$Priority_j(t)$	Time-varying task priority	ε(t)	Active communication links
AssignCost $(\boldsymbol{u_i}, \boldsymbol{t_j})$, PathCost $_{ij}$	Dispatch and path costs	C_{total}	Total mission cost objective
$\boldsymbol{a}_{(i,t)}$, $\boldsymbol{s}_{(i,t)}$, $\boldsymbol{O}_{(i,t)}$	Action, state, observation	$\mu_i\left(s\middle\vert\theta_i^{\mu}\right)$, $Q_i\left(s,a\middle\vert\theta_i^{Q}\right)$	Actor and critic networks
$\boldsymbol{\theta_i^{\mu}}$, $\boldsymbol{\theta_i^{Q}}$	Actor/critic parameters	$\theta_i^{\mu'}$, θ_i^{Q}	Target network parameters
α_a, α_c, τ	Actor/critic learning rates; soft update	γ	Discount factor
R_{comp}, R_{res}	Completion, resource rewards	P_{risk} , P_{fail}	Risk, failure penalties
w1…w6; v1, v2	Reward/penalty weights	$r_{i,t}$, $\overline{R}$	Immediate reward, discounted return
K, $\boldsymbol{T_c}$	MAML inner steps; meta-episodes	α, β	Inner/outer meta learning rates
B, L, G	Batch size, net depth, width	D, $L_i\left(\theta_i^{Q}\right)$	Replay buffer, critic loss

enable fast adaptation, MAML is incorporated during training and meta-trained on a deliberately diverse task distribution. This distribution is constructed by systematically varying key disaster-related factors, including static and dynamic obstacle density and layout, task types and priorities, communication conditions (LoS/NLoS), and initial UAV states such as position and energy. Exposure to this structured variability allows each UAV to adapt quickly to new but structurally similar disaster scenarios using only a few gradient updates at deployment. As with prior meta-learning approaches, performance may degrade strongly out-of-distribution (OOD) scenarios that differ fundamentally from the training conditions. However, RCTP is designed to support rapid near-distribution adaptation, which is most relevant for real disaster response where new scenarios typically share underlying structure with past experiences. Importantly, the additional computational cost of MAML is limited to offline training, while online execution involves only lightweight inference and minimal updates, preserving real-time feasibility on resource-constrained UAV platforms.

Fault tolerance, path replanning, and resource-aware coordination

In dynamic disaster environments, UAVs must operate reliably under limited energy, changing task priorities, and possible failures. RCTP ensures fault tolerance through continuous UAV health monitoring, energy-aware coordination, and real-time task reassignment. Each UAV maintains an operational status vector $F(u) = \{Os_1, \cdots, Os_n\}$, where $Os_i = 0$ indicates UAV failure. When a failure is detected, all tasks assigned to the failed UAV are immediately released. Critical tasks are then reassigned in real time to feasible UAVs with sufficient residual energy, communication connectivity, and capability, using updated priority-aware suitability scores. To preserve mission efficiency during reassignment, RCTP explicitly balances task urgency and energy constraints. Energy-limited UAVs are excluded from energy-intensive tasks, while UAVs with higher residual energy are prioritized. The allocation objective penalizes execution and path costs, ensuring that the reassigned tasks incur minimal additional delay and energy consumption while maintaining high-priority mission objectives. Following reassignment, the affected UAVs perform on-the-fly path replanning by minimizing travel cost while respecting obstacle, no-fly, and safety constraints, $\min \sum_{i=1}^{n} \sum_{j=1}^{m} X_{ij} D_{ij}$, where D_{ij} denotes the distance traveled by UAV u_i to execute task t_j. To ensure smooth trajectory updates after replanning, UAV positions are updated with bounded step size $\rho > 0$, elevation angle $\theta[t] \in [0, \pi]$, and heading $\psi[t] \in [0, 2\pi]$, subject to altitude constraints:

$$\mathrm{p}_i(t+1) = \begin{bmatrix} x_i(t) + \rho \sin\theta[t] \cos\psi[t] \\ y_i(t) + \rho \sin\theta[t] \sin\psi[t] \\ z_i(t) + \rho \cos\theta[t] \end{bmatrix}, \; h_{\min} \le z_i(t) \le h_{\max}, t \in [1, T-1] \tag{1}$$

Residual resources are updated after every assignment to preserve feasibility for subsequent decisions: $R_{\mathrm{rem}}(i) \leftarrow R_{\mathrm{rem}}(i) - \sum_{j=1}^{m} R_j^{\mathrm{req}} X_{ij}, \; i = 1, \ldots, n$, where R_j^{req} denotes the resource requirement of task t_j. The overall objective minimizes mission cost while allowing partial task completion when necessary: $\min C_{\mathrm{total}} = \sum_{i=1}^{n} \sum_{j=1}^{m} \left(\text{AssignCost } \left(u_i, t_j\right) + \text{PathCost }_{ij} \right) X_{ij} + \mu \text{ Penalty }_{\text{uncompleted}}$ with $\mu > 0$ set by mission policy. This formulation explicitly handles failures, regenerates feasible paths, preserves trajectory continuity, and maintains resource awareness throughout execution.

Task priority

Tasks are ranked by urgency and assigned to UAVs that can feasibly execute them. Let $Priority_j(t)$ be the time-varying priority of task t_j. The scheduler maximizes total served priority, $\max \sum_{i=1}^{n} \sum_{j=1}^{m} Priority_j(t) X_{ij}$, subject to resource, connectivity, safety, and capacity constraints, where $X_{ij} \in \{0, 1\}$ indicates assignment of UAV u_i to t_j. Task priorities evolve in real time, increasing with mission urgency and value, and decaying near deadlines to discourage late execution. When full completion is infeasible, partial progress is

evaluated by weighting outcomes with Priority$_j(t)$, ensuring that the objective remains meaningful under resource scarcity. Fig. 2 illustrates the timing of these priority-driven assignments for individual UAVs and coalitions.

Load balancing

To prevent overloading individual UAVs and improve scalability, RCTP explicitly balances task assignments across the fleet. Let $L_i = \sum_{j=1}^{m} \text{task_load}_j X_{ij}$ denote the workload of u_i and $\overline{L} = \frac{1}{n}\sum_{i=1}^{n} L_i$ the fleet average. We encourage fairness by minimizing workload variance $\frac{1}{n}\sum_{i=1}^{n}(L_i - \overline{L})^2$, subject to per-UAV capacity constraints $\sum_{j=1}^{m} \text{task_load}_j X_{ij} \le \text{cap}(i)$. We add a small penalty on rapid workload changes $|L_i^t - L_i^{t-1}|$ to reduce unstable switching when priorities or connectivity change. When two options are nearly equal, we break ties using the suitability score and the current load, ensuring urgent tasks are served first without overloading any single UAV. This strategy promotes balanced utilization, improves robustness under dynamic conditions, and supports scalable multi-UAV coordination.

Coalition formation and suitability score

To handle highly heterogeneous UAV swarms, RCTP assigns UAVs to tasks using a resource-aware suitability score computed for each UAV-task pair (u_i, t_j), as $S_{ij} = w_1 \text{PL}_i + w_2 B_i + w_3 ReqMatch_{ij} - w_4 \widetilde{D}_{ij}$, where PL_i is the payload capacity of u_i, B_i its battery level, $ReqMatch_{ij}$ measures the compatibility between UAV capabilities and task requirements, and $\widetilde{D}_{ij}$ is the normalized UAV-task distance. All terms are min-max normalized to [0,1], and the nonnegative weights $w_1, \ldots, w_4 \ge 0$ satisfy $\sum_{k=1}^{4} w_k = 1$ ensuring interpretability. This formulation directly accounts for differences in UAV energy, sensing capability, payload, and mobility during task allocation. UAVs with limited resources are assigned fewer or less demanding tasks, while more capable UAVs handle high-priority missions. This ensures feasible task assignment and allows the system to scale effectively across diverse UAV types [37]. In RCTP, the suitability score is updated online using the current UAV states, task requirements, and communication conditions. When mission conditions change (e.g., floods or fires), updates in energy levels, task accessibility, and link quality automatically adjust suitability scores, triggering real-time coalition reformation and role reassignment without retraining. The weighting parameters $(w_1, \ldots, w_4)$ are initialized based on mission priorities (e.g., urgency and energy efficiency) and validated via sensitivity analysis. Task allocation is obtained by maximizing a priority- and suitability-aware objective:

$$\max Z = \sum_{i=1}^{n}\sum_{j=1}^{m} X_{ij}\left(\alpha Priority_j + \beta S_{ij}\right) - \lambda \sum_{i=1}^{n}\sum_{j=1}^{m} X_{ij}\left(\text{AssignCost}\ (u_i, t_j) + \text{PathCost}_{ij}\right) \tag{2}$$

where $\alpha, \beta, \lambda > 0$ are selected using the same sensitivity-based procedure. Assignments are subject to energy, capacity, and connectivity constraint, $\sum_{j=1}^{m} \text{EnergyUse}_{ij} X_{ij} \le B_i$, $\sum_{j=1}^{m} \text{task_load}_j X_{ij} \le \text{cap}_i$, *and* $\| \text{p}_i(t) - \text{p}_k(t) \|_2 \le R$, $\forall (i,k) \in \varepsilon(t)$, ensuring feasibility, platform safety, and coalition connectivity during coordinated missions.

System model and problem formulation

RCTP considers a swarm of heterogeneous UAVs $U = \{u_1, u_2, \ldots . u_n\}$, where each UAV differs in payload capacity, speed, sensing

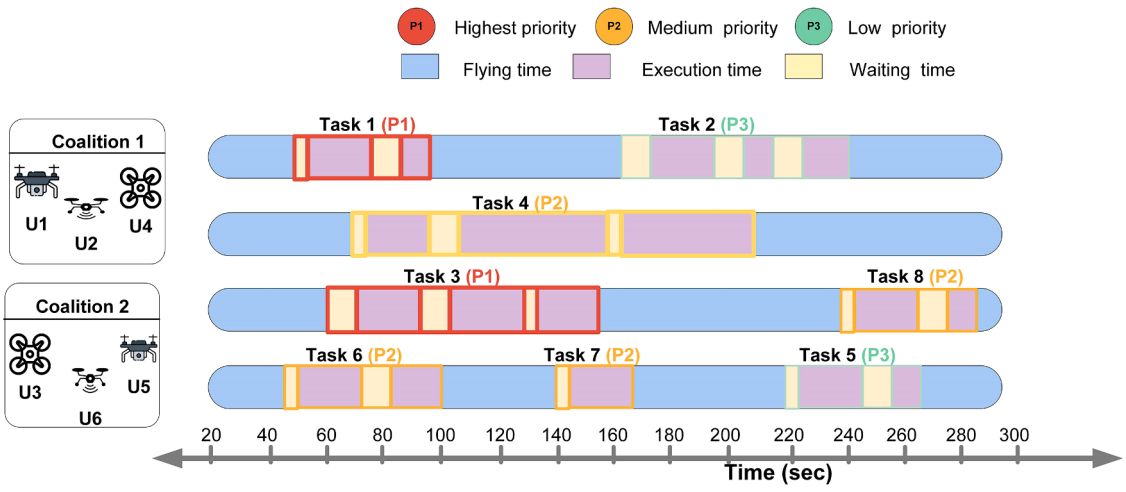


Fig. 2. Task sequences for UAV coalitions with corresponding priority levels.

capability, communication quality, and available energy. The mission consists of a set of tasks $T = \{t_1,\ t_2, \ldots..t_m\}$ distributed in 3D space, each associated with a nominal execution region $A = \{A_1,\ A_2, \ldots..A_m\}$. The operating environment contains both static and dynamic obstacles $Obs = \{Obs_1,\ Obs_2, \ldots..Obs_o\}$, modeled as 3D polygons defined by $[\![x_{obs}, y_{obs},\ h_{obs}]\!]$. Communication among UAVs may be intermittent and subject to LoS/NLoS conditions, resulting in partial and noisy state information. Each UAV is equipped with onboard sensors, GPS, actuators, and wireless radios to perceive the environment, localize itself, execute tasks, and coordinate with neighboring agents. RCTP employs MA-DDPG with decentralized actor policies $\mu_i(.|\theta_i^{\mu})$ that output continuous control actions, and a centralized critic $Q_i\left(.|\theta_i^{Q}\right)$ used during training to facilitate coordinated learning under partial observability. To enable rapid adaptation to new disaster scenarios, the actor-critic parameters are meta-trained using MAML over a diverse distribution of tasks and environments. At deployment, each UAV adapts its policy with only a few gradient updates using limited local experience, enabling fast, scalable, and real-time coordination in unseen and dynamic disaster conditions.

Observation $O_{i,t}$ space

The observation of UAV $u_i \in U$ at time t aggregates local states and shared summaries:

$$O_{i,t} = \underbrace{p_i(t), v_i(t), \psi_i(t), D_i^{\mathrm{tx}}(t)}_{\text{kinematics \& buffer}}, \underbrace{\{\mathrm{Obs}_k\}_{k=1}^{o}}_{\text{obstacles}}, \underbrace{\left\{ \mathrm{task_state}_j(t), a_j, A_j \right\}_{j=1}^{m}}_{\text{tasks \& regions}}, \\ \underbrace{E_i(t), B_i(t), \mathrm{PL}_i}_{\text{energy/battery/payload}}, \ \underbrace{\{D_{ij}(t)\}_{j=1}^{m}}_{\text{UAV-task distances}}, \ \underbrace{\mathrm{link}_i(t), \mathcal{N}_i(t)}_{\text{link quality \& neighbors}} \ \}. \tag{3}$$

Here, $p_i(t)$ and $v_i(t)$ denote the position and velocity of UAV u_i, $\psi_i(t)$ is its heading, and $D_i^{tx}(t)$ represents buffered data to be transmitted. The task state $\mathrm{task_state}_j(t) \in \{\mathrm{pending}, \mathrm{assigned}, \mathrm{done}\}$ and a_j denote the status and location of task t_j. $E_i(t)$, $B_i(t)$, and PL_i represent energy level, battery capacity, and payload/sensor capability, respectively; $D_{ij}(t)$ is the distance between UAV u_i and task t_j; and $\mathrm{link}_i(t)$ summarizes the communication link quality to neighboring UAVs $\mathcal{N}_i(t)$. All quantities are normalized. In realistic disaster environments, task states, inter-UAV distances, and link qualities are noisy, delayed, and incomplete due to NLoS propagation, smoke, debris, interference, and sensing uncertainty. Accordingly, RCTP treats observations as imperfect local estimates rather than precise global states. Robust coordination under such partial observability has been studied in multi-agent reinforcement learning, where agents learn policies from limited and noisy local observations to achieve cooperative behavior without full state information [38,39]. In RCTP, robustness is achieved through three mechanisms: (i) decision-making prioritizes locally sensed information and short-range neighbor exchange, reducing reliance on global knowledge; (ii) communication features explicitly encode the degraded LoS/NLoS link quality, allowing policies to adapt under intermittent connectivity; and (iii) policies are trained under noisy and partially observable conditions during MAML-based meta-training, improving stability and generalization at deployment. As a result, RCTP maintains reliable coordination and task execution even when sensing and communication quality deteriorates.

State space $s_{i,t}$

Each UAV maintains a local state as $s_{i,t} = O_{i,t},\ i \in U, t = 1, 2, \ldots, T$, reflecting partial and noisy observability of the environment. During training, a centralized critic leverages joint state-action summaries to facilitate coordinated learning, while execution remains fully decentralized. MAML supplies meta-initialized actor-critic parameters, enabling rapid online adaptation to change in task distribution, environmental conditions, and communication quality with only a few gradient updates.

Action space $a_{i,t}$

Actions are continuous and include both motion and coordination decisions:

$$a_{i,t} = \{\underbrace{v_i(t), \psi_i(t), \theta_i(t)}_{\text{motion control}}, \underbrace{\pi_i^{\mathrm{task}}(t)}_{\text{task selection}}, \underbrace{\pi_i^{\mathrm{coal}}(t)}_{\text{coalition formation}}, \underbrace{\pi_i^{\mathrm{path}}(t)}_{\text{path update}}, \underbrace{\pi_i^{\mathrm{comm}}(t)}_{\text{comm/sensing mode}} \ \} \tag{4}$$

where $v_i(t)$, $\psi_i(t)$, and $\theta_i(t)$ control the UAV's speed, heading, and climb/descent. $\pi_i^{\mathrm{task}}(t)$ represents a continuous preference over candidate tasks, $\pi_i^{\mathrm{coal}}(t)$ encodes coalition participation strength, $\pi_i^{\mathrm{path}}(t)$ adjusts local waypoints or trajectory curvature for replanning, and $\pi_i^{\mathrm{comm}}(t)$ selects communication or sensing modes (e.g., transmission rate or sensor duty cycle) subject to operational constraints. During execution, a lightweight allocator converts these continuous preferences into discrete task assignments $X_{ij} \in \{0, 1\}$ using task priority, suitability scores, and resource feasibility constraints, enabling coordinated yet fully decentralized decision-making.

Reward function

Each UAV maximizes a discounted return $R = \sum_{t=0}^{T} \gamma^t r_{i,t}$, $0 < \gamma \leq 1$, using reward terms normalized to $[0, 1]$ for interpretable weighting across scenarios. The positive components are: (i) task completion, $R_{\mathrm{comp}} = \sum_j Priority_j(t) 1\{\mathrm{task}\ t_j\ \mathrm{completed}\}$, which

credits priority-weighted completions and captures partial mission value; (ii) suitability match, $R_{\text{suite}} = \overline{S}_{ij}$, encouraging assignments aligned with UAV capabilities; (iii) resource efficiency, $R_{\text{res}} = \frac{1}{(1+C_{\text{res}})}$, where $C_{\text{res}} = \alpha\widehat{\Delta t} + \beta\widehat{\Delta e}$ with $\alpha, \beta \geq 0$, promoting time and energy-efficient behavior; (iv) fault recovery, $R_{\text{fault}} = \delta 1\{\text{recovered}\}$, $\delta > 0$, rewarding successful reassignment after failures; (v) path quality, $R_{\text{path}} = 1/(1 + \overline{\text{travel_time}})$; and (vi) coalition efficiency, $R_{\text{coal}} = \varepsilon\,\overline{\text{collab_efficiency}}$ with $\varepsilon > 0$. Penalties include: risk, $P_{\text{risk}} = \rho\,\overline{\text{risk_level}}$, capturing separation violations, no-fly incursions, excessive energy draw, and link loss, and failure, $\eta \text{Priority}_j(t)F_j(t),\ \eta \geq 0$. where $F_j(t) = 1$ if task t_j fails or times out. The instantaneous reward is $r_{i,t} = \sum_{k=1}^{6} w_k R_k - \sum_{\ell=1}^{2} \nu_\ell P_\ell$, with non-negative weights w_k and ν_ℓ. All terms are computed from local observations and shared summaries, keeping the online cost low while aligning individual behavior with mission objectives under resource constraints and uncertainty.

Actor-Critic updates

RCTP adopts MA-DDPG with decentralized deterministic actors and a centralized critic to enable coordinated learning under partial observability. For agent i, the deterministic policy gradient is

$$\nabla_{\theta_i^\mu} J(\mu_i) = \mathbb{E}_{(s,a)\sim\mathscr{D}}\left[\nabla_{\theta_i^\mu}\mu_i(s)\nabla_a Q_i\left(s, a; \theta_i^Q\right)\Big|_{a=\mu_i(s)}\right], \tag{5}$$

where $\mathscr{D}$ is the replay buffer, and Q_i is the critic for agent i. Each agent maintains an actor μ_i, a critic Q_i, target networks $\mu_i^{'}, Q_i^{'}$, and a local replay buffer. The critic is trained by minimizing the temporal-difference (TD) loss

$$\mathscr{L}\left(\theta_i^Q\right) = \mathbb{E}_{(s,a,r,s')\sim\mathscr{D}}\left[\left(Q_i\left(s, a; \theta_i^Q\right) - y\right)^2\right],\ y = r + \gamma Q_i^{'}\left(s', \mu_i^{'}(s')\right), \tag{6}$$

with discount $\gamma \in (0, 1]$. The actor is then updated using the deterministic policy gradient

$$\nabla_{\theta_i^\mu} J\left(\theta_i^\mu\right) = \mathbb{E}_{s\sim\mathscr{D}}\left[\nabla_{\theta_i^\mu}\mu_i(s)\nabla_a Q_i\left(s, a; \theta_i^Q\right)\Big|_{a=\mu_i(s)}\right]. \tag{7}$$

Target networks are softly updated to improve training stability:

$$\theta_i^{Q'} \leftarrow \tau\theta_i^Q + (1-\tau)\theta_i^{Q'},\ \theta_i^{\mu'} \leftarrow \tau\theta_i^\mu + (1-\tau)\theta_i^{\mu'},\ \tau \in [0, 1]. \tag{8}$$

To enable rapid adaptation to new disaster scenarios, the actor-critic parameters are meta-trained using MAML over a distribution of tasks $\{T_j\}$. For each task T_j, inner-loop adaptation is performed as

$$\theta_i^{'} = \theta_i - \alpha\nabla_{\theta_i}\mathscr{L}_{\mathscr{T}_j}(\theta_i), \tag{9}$$

followed by the outer-loop meta-update:

$$\theta_i \leftarrow \theta_i - \beta\sum_{\mathscr{T}_j}\nabla_{\theta_i}\mathscr{L}_{\mathscr{T}_j}\left(\theta_i^{'}\right), \tag{10}$$

with inner/outer learning rates $\alpha, \beta > 0$. This yields meta-initialized actors/critics that adapt with few samples while preserving continuous-control fidelity for coalition formation, tasking, and path planning.

Estimation of collision probability

Accurate collision risk estimation is essential for UAV safety in cluttered, uncertain environments. We model relative position error with a zero-mean Gaussian whose covariance C_e, whose probability density function is

$$f\left(e_{ij}\right) = \frac{1}{(2\pi)^{\frac{3}{2}}|C_e|^{\frac{1}{2}}}\exp\left(-\frac{1}{2}\mathrm{d}^\top C_e^{-1} e_{ij}\right), \tag{11}$$

where e_{ij} denotes the relative displacement error. The collision probability at time t is the probability mass d over the overlap volume V_{overlap} of the two safety envelopes:

$$P_{\text{coll}}(t) = \iiint_{V_{\text{overapp}}} f(d)\,dd, \tag{12}$$

where V_{overlap} is the intersection of the spherical safety regions centered at the two bodies. For isotropic covariance and spherical envelopes, closed-form approximations are used; otherwise, the integral is estimated through Monte Carlo sampling. This probabilistic formulation allows collision risk to be incorporated into safety constraints and reward penalties under sensing uncertainty and noisy state estimates.

Collision-Free paths

The Euclidean separation between UAVs i and j at time t is

$$D_{ij}(t) = \sqrt{\left(x_i - x_j\right)^2 + \left(y_i - y_j\right)^2 + \left(z_i - z_j\right)^2} \tag{13}$$

$D_{ij}(t) \geq d_{\min} \forall (i,j), t$, where d_{min} is the minimum distance required to avoid collision. UAVs must adjust their speed or altitude if the distance between them is less than d_{min} to avoid collisions.

Channel model

We model LoS and NLoS links to capture realistic wireless conditions in UAV swarms. Following a probabilistic elevation-angle model, the LoS probability between two UAVs with elevation angle θ is

$$P_{LoS}(\theta) = b_1 \left(\frac{180}{\pi}\theta - \zeta\right)^{b_2}, P_{NLoS}(\theta) = 1 - P_{LoS}(\theta) \tag{14}$$

where b_1, b_2, ζ are environment dependent parameters. LoS path loss follows free-space propagation,

$$L_{LoS}(dB) = 20\log_{10}(d) + 20\log_{10}(f) + 20\log_{10}\left(\frac{4\pi}{c}\right) \tag{15}$$

while NLoS links incur additional attenuation η_{NLoS} due to diffraction and scattering:

$$L_{NLoS}(dB) = L_{LoS}(dB) + \eta_{NLoS} \tag{16}$$

The expected path loss is: $\overline{L}(\text{dB}) = P_{\text{LoS}}(\theta) L_{\text{LoS}} + P_{\text{NLoS}}(\theta) L_{\text{NLoS}}$. Given bandwidth B_i, transmit power P_i^{tx}, separation D_{ij}, path-loss exponent α_p, and noise spectral density N_0, the achievable rate is $R_i = B_i \log_2\left(1 + \frac{P_i^{\text{tx}} D_{i,j}^{-\alpha_p}}{B_i N_0}\right)$. These channel metrics are incorporated as state features and constraints, enabling the policy to adapt transmission power, bandwidth allocation, and link selection under time-varying LoS/NLoS conditions.

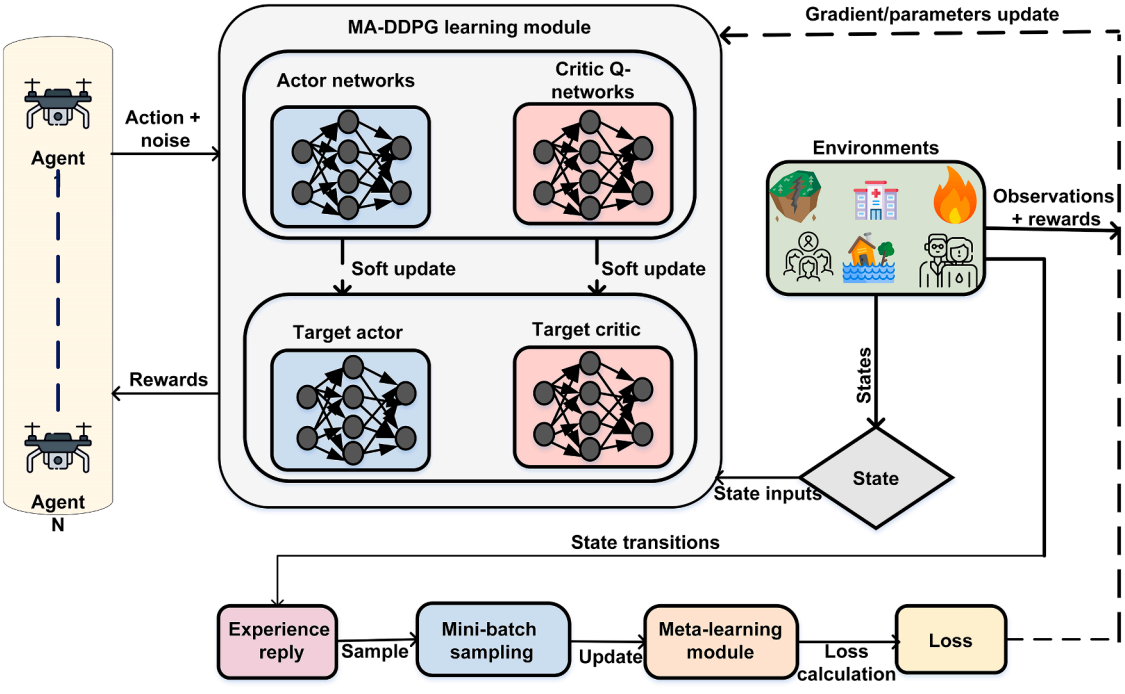


Fig. 3. MAML-based MA-DDPG framework for coalition formation, task assignment, and path planning in RCTP.

Algorithm 1
MAML enhanced resource-aware coalition formation, dynamic task assignment, and path planning for multiple UAVs in RCTP.

Input: Initial values of α_a, α_c, $\theta^{Q'}$, $\theta^{\mu'}$,γ, τ, $\{\theta_i^{\mu}, \theta_i^{Q}\}_{i=1}^{n}$, $\{\theta_i^{u'}, \theta_i^{Q'}\}_{i-1}^{n}$, N_i, α, β, $\{\mathcal{T}_j\}$, $d_{\min}$, environment maps and task set $\mathcal{G}$
Output: C_{best}, p_{UAV_i} and U_T' for each UAV
//Initialize environments and models
Initialize disaster maps, obstacles, threat layers, task set $\mathcal{G}$.
For each UAV u_1 , encode initial observation $O_{i,0} \rightarrow s_{i,0}$.
Randomly initialize actor $\mu_i(\cdot \mid \theta_i^{\mu})$and critic $Q_i(\cdot \mid \theta_i^{Q})$.
Set targets: $\theta_i^{\mu'} \leftarrow \theta_i^{\mu}, \theta_i^{Q'} \leftarrow \theta_i^{Q}$.
Create replay buffer $D \leftarrow \emptyset$; set $C_{\text{bast}} \leftarrow \emptyset, \{p_{m_1}\} \leftarrow \emptyset, U_T' \leftarrow \emptyset, u^* \leftarrow -\infty$.
Define task-specific environments T_j for meta-training.
// Meta-training phase (learn fast-adapt initialization)
for each meta-episode **do**
  for each UAV i and sampled task T_j **do**
    Compute task loss $\mathcal{L}_{T_j}(\theta_i)$
    Inner loop updates as in (9)
  end for
  Outer loop updates as in (10)
end for
// Environment interaction and learning
for each episode **do**
  Reset env if needed, refresh tasks/maps.
  for each UAV i **do**
    Receive initial observation $O_{i,t}$ and derive initial state $s_{i,t}$
    for t=1 to T **do**
      Select action$a_{i,t} \leftarrow \mu_1(s_{i,t} \mid \theta_i^{\mu}) + N_t$
      Execute $a_{i,t}$; observe $O_{i,t+1} \rightarrow s_{i,t+1}$
      Compute immediate reward $r_{i,t}$ **(Algorithm 2)**
      Store transition $(s_{i,t}, \alpha_{i,t}, r_{i,t}, s_{i,t+1}) \in D$.
      If unsafe ($D_{\min}^{(i,t)} < d_{\min}$, link loss, no-fly) then abort path and update observation.
    end for
//Policy and value update (mini-batch off-policy MA-DDPG)
  Sample mini-batch $B \subset R$.
  $y \leftarrow r + \gamma Q_i'\left(s', \mu_1'(s') \mid \theta_i^{Q'}\right)$
  Critic step: $\theta_i^{Q} \leftarrow \theta_i^{Q} - \alpha_c \nabla_{\theta_1^{Q}} \frac{1}{|B|} \sum_B \left(Q_i(s, a \mid \theta_i^{Q}) - y\right)^2$
  Actor step: $\theta_i^{\mu} \leftarrow \theta_i^{\mu} + \alpha_a \frac{1}{|B|} \sum_B \left[\nabla_{\theta_i^{\mu}} \mu_i(s) \nabla_a Q_i(s, a \mid \theta_i^{Q})\right]_{a-\mu_4(a)}$
  Soft targets: $\theta_i^{Q'} \leftarrow \tau\theta_i^{Q} + (1-\tau)\theta_i^{Q'}; \theta_i^{\mu'} \leftarrow \tau\theta_i^{\mu} + (1-\tau)\theta_i^{\mu}$.
end for
//Operational planning (uses updated networks)
Compute priorities $P_i(t)$, suitability S_{ij}, and feasibility (energy, payload, links, capacity).
Form coalitions: $C_{\text{curr}} \leftarrow$ FormCoalitions $(\{u_i\}, \{t_j\}, S_{ij}$, constraints).
Assign tasks urgent-first with load balance: $X \leftarrow$ AssignTasksUrgentFirst $(C_{\text{curr}}, P, S$, constraints).
Replan paths enforcing obstacles, no-fly, connectivity, d_{min}: $\{p_{m_i}\} \leftarrow$ ReplanPaths ($\{u_i\}, X$, maps).
Evaluate mission utility u_{curr} (priority-weighted completion, energy, risk, fairness).
if $u_{\text{curr}} > u^*$, set **then**
  set $(C_{best}, p_{UAV_i}, U_T'') \leftarrow (C_{\text{curr}}, \{p_{m_i}\}, X)$ and $u^* \leftarrow u_{\text{curr}}$.
end if
end for
//Evaluation and adaptation
Regularly evaluate TCR, fairness, energy per task, collision risk, link uptime, fault-recovery time.
If required, repeat meta-adaptation with new $T_j$to maintain responsiveness in evolving environments.
return C_{best}, p_{UAV_i} and U_T'

Proposed RCTP algorithm

The proposed RCTP algorithm enables UAVs to adapt quickly to changing disaster environments by learning a flexible policy that can be fine-tuned with only a small amount of new data. During training, UAVs practice coalition formation, task assignment, and safe path planning across diverse simulated missions. Each agent performs a few gradient updates per task, and the system learns a shared meta-policy that is refined with minimal data at deployment. Learning is guided by rewards aligned with mission goals, including priority-weighted task completion, efficient resource use, and risk-aware navigation. As shown in Fig. 3, this integration supports fast, coordinated, and intelligent decision making by multi-UAV teams in real-time disaster response.

Algorithm 1 combines MAML with MA-DDPG to train UAV swarms that can form coalitions, assign tasks, and plan paths adaptively and resource-efficiently in dynamic disaster settings. In meta-training, UAVs practice diverse simulated missions to learn a shared initialization that adapts to new scenarios with a few gradient steps. During execution, each UAV observes its state, selects continuous actions, receives a mission-aligned reward, and updates its actor-critic with soft target updates. A planning layer uses suitability scores to form coalitions, assigns urgent tasks subject to feasibility and load-balancing, and replans paths to maintain separation, avoid no-fly zones, and preserve connectivity; detected failures or link losses trigger immediate reassignment and replanning. The system tracks mission utility and retains the best coalition plan and paths.

Algorithm 2 gives pseudocode to compute the immediate total reward $r_{i,t}$ for each UAV during task execution in the RCTP framework. At each step, the UAV's action is scored on priority-weighted task completion, time and energy efficiency, fault recovery, path quality, and coalition effectiveness. Penalties are added for task failure and safety risks such as entering no-fly zones, collisions, or link loss, and other inefficient choices. The final reward is a weighted sum of the positive terms minus the penalties. This single value guides learning toward safe, efficient, and cooperative behavior in dynamic disaster environments.

Algorithm 2
Immediate reward calculation.

Input: u_i, $s_{i,t}$, $a_{i,t}$, $\{t_j\}$, coalition info, $d_{\min}$; $\{w_k, v_\ell\}$, $\alpha_t, \beta_e, \delta, \eta, \rho, \varepsilon$
Output: $r_{i,t}$
Initialize $r_{i,t} \leftarrow 0$
// Safety and link checks
Compute distances to tasks.
$d_{\min} \leftarrow \min_{k \neq i} D_{ik}(t)$
$P_{\text{coll}} \leftarrow$ EstimateCollisionProbability (u_i, neighbors, obstacles)
$in_nofly \leftarrow InNoFlyZone\ (u_i)$
$link_{ok} \leftarrow RequiredLinksAvailable\ (u_i)$
//Reward components
$R_{\text{comp}} \leftarrow P_{j^*}(t) \cdot 1$ {completed t_{j^*} }
Resource cost $C_{\text{res}} \leftarrow \alpha_t \widehat{\Delta t} + \beta_e \widehat{\Delta e}$
$R_{\text{res}} \leftarrow 1/\left(1 + C_{\text{res}}\right)$
$R_{\text{fault}} \leftarrow \delta \cdot 1$ {fault recovered}
$R_{\text{path}} \leftarrow 1/(1 + \text{travel_time})$
$R_{\text{coal}} \leftarrow \varepsilon \cdot$ collab_efficiency
$R_{\text{suite}} \leftarrow \hat{S}_{ij^*}$normalized suitability for $(u_i - t_{j^*})$
//Failure and risk handling
$P_{\text{fail}} \leftarrow \eta P_{j^*}(t) \cdot 1$ {failed or timed out}
$\text{risk_level}_{i,t} = \frac{\lambda_1 1\{D_{\min}^{(i,t)} < d_{\min}\} + \lambda_2 \hat{p}_{\text{coll}}^{(i,t)} + \lambda_3 1\{\text{in_nofly}\} + \lambda_4 1\{\neg\ \text{link_ok}\}}{\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4}, \lambda_k \geq 0$
$P_{\text{risk}} \leftarrow \rho \cdot \text{risk_level}_{i,t}$
if ($D_{\min}^{(i,t)} < d_{\min}$) or in_nofly or $link_ok$ **then**
Abort the path coordinates.
end if
Calculate total immediate reward as: $r_{i,t} = \sum_{k=1}^{6} w_k R_k - \sum_{\ell=1}^{2} v_\ell P_\ell$
return $r_{i,t}$

Complexity analysis

RCTP operates in two stages: offline meta-training and online execution, which enables practical deployment on resource-constrained UAV platforms. Offline training integrates MAML with MA-DDPG and is performed on ground/cloud infrastructure. For N UAVs, M tasks, K inner steps, batch size B, T_c meta-episodes, and actor/critic networks of depth L and width G, the total offline complexity is $\mathcal{O}(T_c MKBNLG^2)$, which is incurred only during training and does not affect onboard operation. Online execution is designed to be lightweight and suitable for embedded flight controllers. At each control step, the dominant costs include: (i) batched actor inference and critic evaluation with complexity $O(NLG^2)$. (ii) coalition scoring and task assignment with worst-case $O(NM)$, reduced in practice to near-linear complexity via top-k pruning (typically $3 \le k \ll N$). (iii) event-triggered replanning for a subset of agents, $O(n_{\text{replan}}\ C_{\text{plan}}\ )$; (iv) safety and connectivity checks over local neighborhoods, $O(Nk)$. The complexity of optional online MA-DDPG update is $O(BNLG^2)$, which can be executed asynchronously or disabled during latency-critical flight phases. Thus, the dominant per-step online complexity is $O(\ (1\ +\ B)NLG^2\ +\ NM\ +\ n_{\text{replan}}\ C_{\text{plan}}\ +\ Nk)$. Over S control steps, the total end-to-end complexity is

$$\mathcal{O}(T_c MKBNLG^2) + O(S[(1+B)NLG^2 + NM + n_{replan}\ C_{plan}\ + Nk]). \tag{17}$$

In practice, online execution mainly involves forward passes through small neural networks and limited interaction with nearby UAVs. Experiments show that the average inference time is below 20 *ms* per control step on an ARM-based embedded UAV processor, which meets real-time flight requirements. The computationally expensive meta-training is performed offline, while online execution remains lightweight [40]. This clear separation ensures that RCTP is scalable, responsive, and practical for deployment on resource-constrained UAVs in time-critical disaster scenarios.

Performance evaluation

In this section, we evaluate RCTP via extensive simulations and compare it against three widely used state-of-the-art RL baselines for task assignment and path planning; we organize the discussion into three subsections: performance metrics, simulation environment, simulation results & discussion.

Performance metrics

1. Task Completion Rate (TCR)We measure the effectiveness of algorithms by the fraction of tasks finished:

$$\text{TCR} = \frac{\text{Task}_{\text{comp}}}{\text{Task}_{\text{total}}} \tag{18}$$

2. Fairness

We assess how evenly workload/resources are distributed across UAVs using Jain's fairness index:

$$f = \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n\sum_{i=1}^{n} x_i^2} \tag{19}$$

where x_i is the workload (or allocated resource) for UAV i. Values closer to 1 indicate more balanced assignments.

3. Energy Consumption Model

RCTP accounts for the main flight phases (i.e., acceleration/translation and hovering) and then integrates power over time to obtain total energy. Let $P_i^{\text{flight}}(v)$ be the instantaneous propulsion power of UAV i at airspeed v. A common rotorcraft model combines profile, induced, and parasitic components:

$$P_{\text{i}}^{\text{flight}}(v) = P_0\left(1 + \frac{3v^2}{U_{\text{tip}}^2}\right) + P_1\left(\sqrt{1 + \frac{\rho^2 A^2 v^4}{W^2}} - \frac{\rho A v^2}{W}\right) + \frac{1}{2} d_0 \rho s A v^3 \tag{20}$$

where P_0 and P_1 are the blade profile and induced power terms, respectively, U_{tip} is blade-tip speed, ρ is air density, A is rotor disc area, s is rotor solidity, d_0 is zero-lift drag coefficient, and W is vehicle weight. During hover or near-hover task execution, power is dominated by induced effects:

$$P_{\text{i}}^{\text{hover}} = P_{\text{i}}\sqrt{1 + \left(\frac{\text{aspP}}{2W}\right)^2} \tag{21}$$

where SFP is the fuselage-equivalent flat-plate area and v is the (small) airspeed during station-keeping. Total mission energy for UAV i integrates power over the flight and hover intervals:

$$E_i^{\text{total}} = \int_0^{t_1} P_i^{\text{flight}}(t)dt + \int_{t_1}^{t_2} P_i^{\text{hover}}(t)dt \tag{22}$$

where t_1 is time-to-task (translation) and t_2 is mission end (including hover/task time). The swarm's average energy is

$$E_{\text{Avg}} = \alpha_1 \frac{1}{N} \sum_{i=1}^{N} E_i^{\text{total}} \tag{23}$$

where α_1 refers to the cost coefficient, which is utilized to convert parameters such as battery life and speed.

4. Average Reward

We report the mean per-step reward across agents and evaluation episodes:

$$\overline{R} = \frac{1}{ETN} \sum_{e=1}^{E} \sum_{t=1}^{T} \sum_{i=1}^{N} r_{i,t} \tag{24}$$

where E is the number of evaluation episodes, T is the steps per episode, N is the number of UAVs, and $r_{i,t}$ is the immediate reward of agent i at step t in episode e.

5. Task Execution Delay

Average delay from assignment to completion across completed tasks:

$$\overline{D}_{\text{taak}} = \frac{1}{|\mathscr{C}|} \sum_{j \in \mathscr{C}} \left(t_j^{\text{comp}} - t_j^{\text{assign}} \right) \tag{25}$$

where $\mathscr{C}$ is the set of completed tasks, t_j^{assign} is the assignment time, and t_j^{comp} the completion time.

6. Average Number of Collisions

We count collision events when separation violates the safety threshold, $D_{\min}^{(t,t)} < d_{\min}$:

$$\overline{C} = \frac{1}{E} \sum_{e=1}^{E} C^{(e)}, \ \overline{C}_{\text{rate}} = \frac{1}{E} \sum_{e=1}^{E} \frac{10^3 C^{(e)}}{T^{(e)}} \tag{26}$$

where $C^{(e)}$ is the total number of collisions in episode e and $T^{(e)}$ its length in steps.

Simulation environment

All simulations were conducted in MATLAB R2023b using the Reinforcement Learning Toolbox on a workstation with an Intel Core

Table 3
Simulation parameters.

Parameter	Value	Parameter	Value
Simulator	MATLAB R2023b (Reinforcement learning toolbox)	U_{tip}; timestamp	118 m/s; 0.1s
Simulation environment	3D (1000 m × 1000 m × 1000 m)	Air density	1.225 kg/m3
UAV's altitude range	(50− 250) m	Blade profile power p_0	99.66 W
Transmission radius for UAV	(250 − 300) m	Induced power p_1	120.16 W
Velocity of UAV	(20 − 40) m/s	Fuselage drag ratio,f_d	0.48
UAV battery capacity	(101- 160) watt hours	Experience buffer length	50,000
Antenna orientation	Omni directional	Discount factor (γ)	0.95
Transmission power	1W	α_a and α_c	0.0001
Propagation channel	Free space	Exploration ratio ($\in$)	0.1
Path loss (LoS)	1 dB	Gradient decay	0.9
Number of UAVs	10–30	Max episodes	2500
Number of tasks	6–20	Minimum distance d_{min}	35 m
Batch size; disk rotor area	64; 0.5 m^2	Episode length	300

i9 CPU, 32 GB RAM, and an NVIDIA RTX 3080 Ti GPU. Both the actor and critic networks use four fully connected layers (400–300–200–200 units) shared across agents with decentralized execution. UAV agents interact with a dynamic 3D environment of size $1000 \times 1000 \times 1000\,\mathrm{m}^3$, containing heterogeneous UAVs with varying payload, sensing, and energy capacities, as well as heterogeneous tasks characterized by location, priority, and resource requirements. The environment includes both static obstacles and dynamic obstacles, requiring continuous path replanning and collision avoidance. To enable rapid adaptation, MAML is applied during training. Meta-training samples diverse task scenarios by varying obstacle density, task distributions, UAV initial states, and communication conditions. Meta-testing evaluates performance on previously unseen task configurations, assessing generalization and few-shot adaptability. The Adam optimizer is used for both inner and outer optimization loops [41]. Exploration in continuous action spaces is achieved using Ornstein-Uhlenbeck noise with parameters $\theta \approx 0.15$and $\sigma \approx 0.2$. Scalability is evaluated by progressively increasing the number of UAVs, tasks, and obstacles, which raises coordination, communication, and replanning complexity. Inter-UAV communication assumes omnidirectional antennas and partial observability, reflecting realistic decentralized operation. The combination of heterogeneous agents, mixed obstacle dynamics, and varying task priorities provides a challenging testbed for evaluating adaptability, robustness, and scalability. All simulation parameters are summarized in Table 3.

Simulation results and discussion

Fig. 4 illustrates the end-to-end execution of the proposed RCTP framework, from UAV deployment to task assignment and obstacle-aware path planning in a 3D environment. In Fig. 4(a), UAVs and tasks are initially placed at random locations. Fig. 4(b) shows the task assignment result, where each UAV (or coalition) is associated with a specific task based on resource availability and task priority. Fig. 4(c) shows the execution phase, in which UAVs navigate toward their assigned tasks while actively avoiding both static and dynamic obstacles, where static obstacles are represented by black cubes while dynamic obstacles are shown as orange ellipses. As these obstacles move, RCTP continuously replans UAV trajectories to maintain safe separation, avoid collisions, and respect no-fly constraints as obstacles move or new hazards appear. The resulting trajectories illustrate energy-aware, collision-free navigation toward assigned tasks. A task is considered completed when a UAV reaches the corresponding task location, as shown in Fig. 4(c).

Fig. 5 illustrates the average reward over 2500 episodes for four different algorithms. It clearly shows that the RCTP algorithm consistently outperforms MA-DDPG, PPO, and DQN in terms of reward collection. Specifically, RCTP improves rewards by 28.57% over MA-DDPG, 50% over PPO, and 60.71% over DQN, respectively. This performance is significant because it shows that RCTP is remarkably good at forming a coalition of UAVs, handling operations and executing tasks based on their importance and difficulty. The distinct advantage of RCTP lies in its systematic and adaptive approach to task handling, which contributes to higher performance compared to other tested algorithms.

Fig. 6 shows how the RCTP algorithm performs in energy efficiency compared to MA-DDPG, DQN, and PPO across three scenarios: increasing the number of UAVs, tasks, and obstacles. As shown in Fig. 8(a), as the number of UAVs increases, RCTP not only keeps its energy efficiency but slightly improves it, performing about 8% better than MA-DDPG. In Fig. 8(b), when the number of tasks grows, all algorithms show a reduction in energy efficiency, but RCTP shows 8% to 10% more efficient than MA-DDPG, 12% to 15% more than DQN, and 15% to 18% more than PPO, respectively. Finally, in the case of increasing obstacles in Fig. 8(c), the RCTP continues to perform better, maintaining energy efficiency that is 10% to 12% higher than MA-DDPG, 15% to 18% more than DQN, and 18% to 20% more than PPO, respectively. Although all algorithms show a slight drop in energy efficiency with more obstacles, RCTP remains the most stable due to its delay minimization, efficient task sharing, collision avoidance, and fault tolerance.

Fig. 7 shows the zero-shot and few-shot adaptation performance of RCTP in an unseen disaster scenario. At zero-shot deployment (adaptation step 0), both MA-DDPG and RCTP achieve similar performance, indicating that initial robustness mainly comes from training on diverse scenarios. However, after only a few gradient updates, the MAML-based RCTP improves rapidly, while MA-DDPG without meta-learning exhibits only marginal improvement. This performance gap clearly indicates that the gain is due to MAML's fast adaptation capability rather than domain randomization alone. The results demonstrate that RCTP can efficiently adapt online to novel disaster environments with minimal additional computation, which is essential for time-critical UAV disaster response.

Fig. 8 compares how quickly RCTP, MA-DDPG, DQN, and PPO algorithms find the best solutions for over 2500 episodes. RCTP demonstrates the fastest convergence time, significantly outperforming the other algorithms. Specifically, RCTP reduces the time it takes to reach optimal solutions by 28.5% compared to MA-DDPG, 42.86% compared to DQN, and 50% compared to PPO, respectively. This improvement shows that RCTP is particularly efficient at quickly finding the best ways to allocate tasks, plan path, and form teams among the UAVs. Throughout the episodes, RCTP demonstrates its strong performance and reliability in dealing with changing and challenging environments. This makes RCTP an excellent choice for scenarios that require quick adaptation and efficient decision-making.

Fig. 9 analyzes how quickly tasks are completed under various conditions using four different reinforcement learning algorithms: RCTP, MA-DDPG, DQN, and PPO. In Fig. 9(a), as more UAVs are introduced, RCTP significantly reduces task completion times by 30% to 40% compared to DQN and PPO. Fig. 9(b) and (c) show that, even as tasks and obstacles increase, RCTP maintains the shortest average task execution delays, achieving 25% to 35% better performance than MA-DDPG and 40% to 50% better than both PPO and DQN, respectively. These results highlight RCTP's ability to effectively scale with the increased number of UAVs and manage more complex scenarios efficiently.

Fig. 10 illustrates the Fairness increases with training for all algorithms. RCTP stays highest throughout and ends 8.7% above MA-DDPG, 16.3% above DQN, and 20.5% above PPO at 2500 episodes. The advantage comes from resource-aware load balancing, priority- and capability-based assignment, coalition formation that spreads work across UAVs, and a MAML-initialized MA-DDPG policy that adapts to changing loads. These mechanisms prevent overload and yield a more even distribution of tasks and resources.

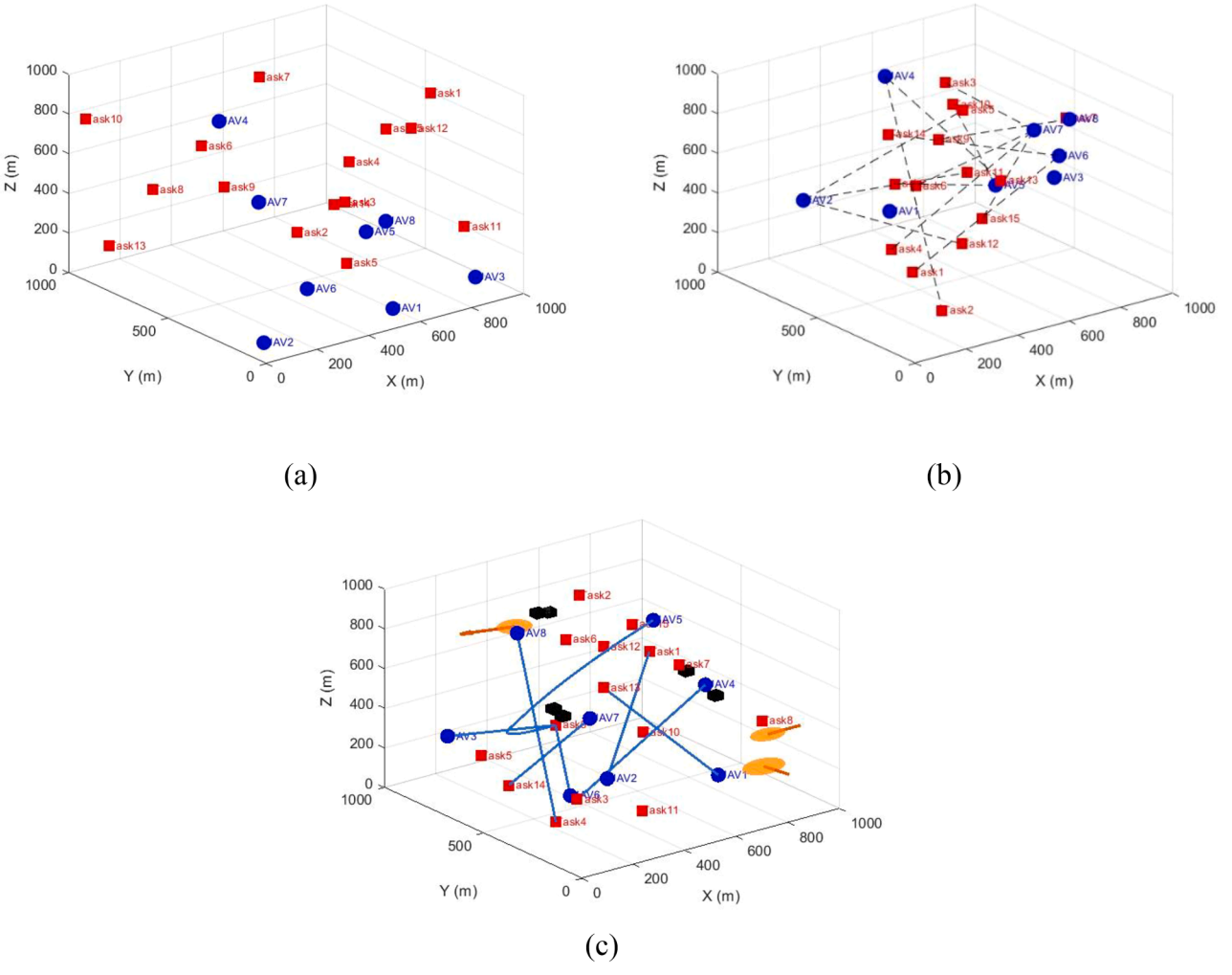


Fig. 4. Implementation scenarios of RCTP: (a) random deployment of UAVs and tasks in 3D space, (b) assignment of tasks to UAVs, and (c) path of UAVs to their respective task location in the presence of static and dynamic obstacles.

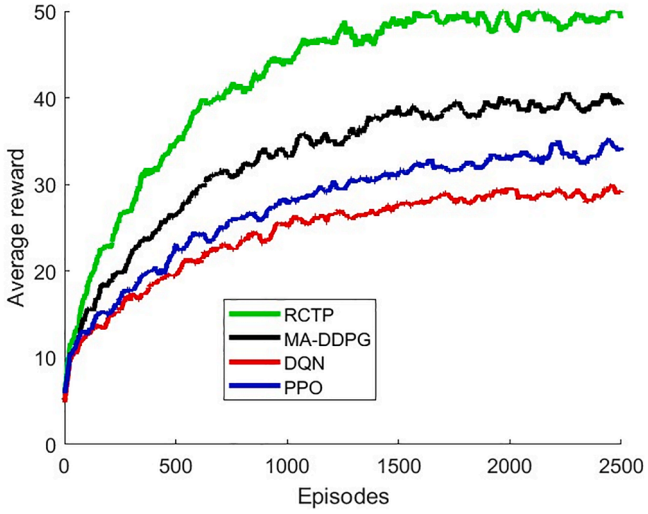


Fig. 5. Average reward versus episodes.

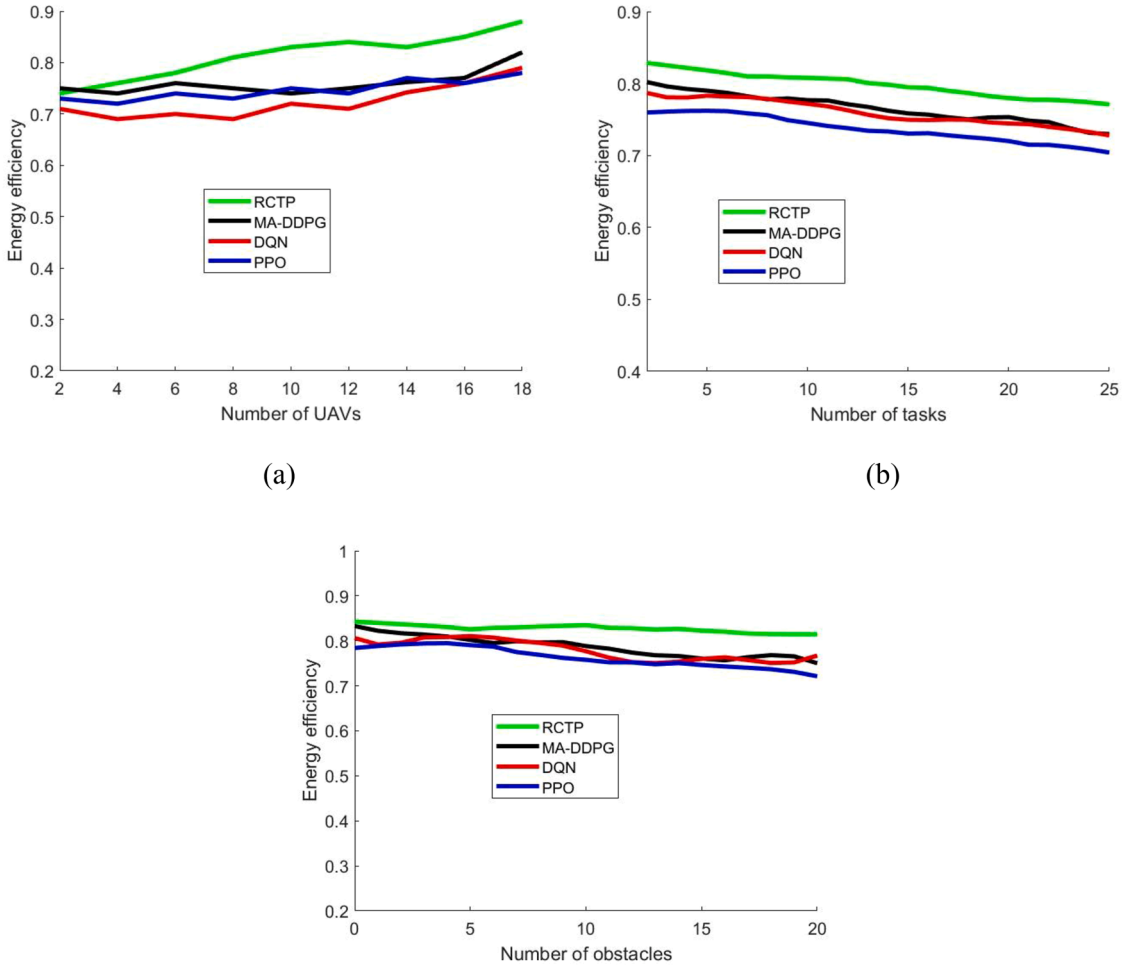


Fig. 6. Energy efficiency: (a) varying the number of UAV, (b) varying the number of tasks, and (c) varying the number of obstacles.

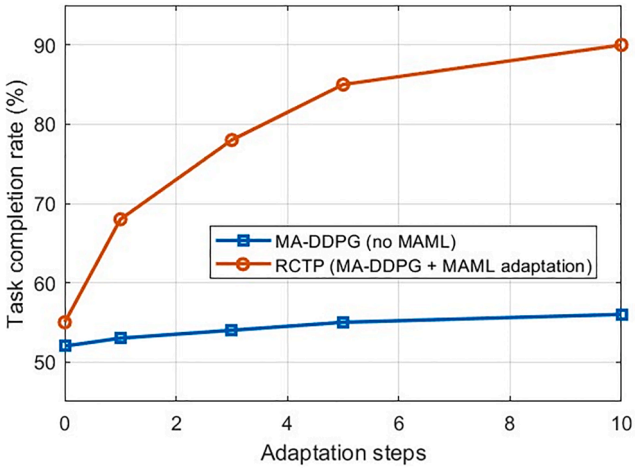


Fig. 7. Zero-shot and few-shot adaptation performance of RCTP.

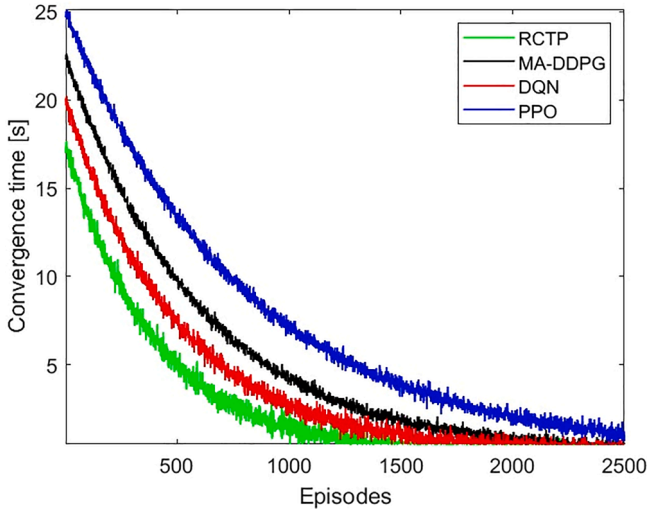


Fig. 8. Convergence time versus episodes.

Fig. 11 compares how well the RCTP, MA-DDPG, DQN, and PPO algorithms perform in reducing collisions. The RCTP algorithm consistently outperforms other algorithms, demonstrating a significantly lower number of collisions in all scenarios. In Fig. 11(a), the RCTP achieves a reduction in collisions of approximately 40% to 50% as the number of UAVs increases, 35% to 45% fewer collisions as the number of tasks increases as shown in Fig. 11(b), and 30% to 40% fewer collisions as the number of obstacles increases as in Fig. 11 (c). These improvements demonstrate the effectiveness of the RCTP in reducing collisions, largely due to better communication and coordination among UAVs and their coalitions. Additionally, the RCTP's effective handling of position prediction errors significantly helps in lowering collision risks.

Fig. 12 presents the ablation study results of RCTP. Removing the resource-aware coalition formation causes the largest performance drop because UAV heterogeneity is no longer considered, leading to inefficient task-UAV assignments. Eliminating MAML-based adaptation reduces the ability to rapidly adapt to new disaster scenarios. Disabling the failure handling module significantly degrades performance when UAV failures occur, since tasks cannot be reassigned efficiently. Finally, removing energy-aware planning results in gradual performance degradation due to inefficient trajectories and higher energy consumption. Overall, the ablation results show that each component contributes a clear benefit, and together they drive the robustness, adaptability, and efficiency of the full RCTP framework.

Discussion and practical insights

The simulation results show that RCTP consistently outperforms MA-DDPG, PPO, and DQN across key metrics, including cumulative reward, task completion rate, convergence speed, energy efficiency, execution delay, fairness, and collision avoidance.

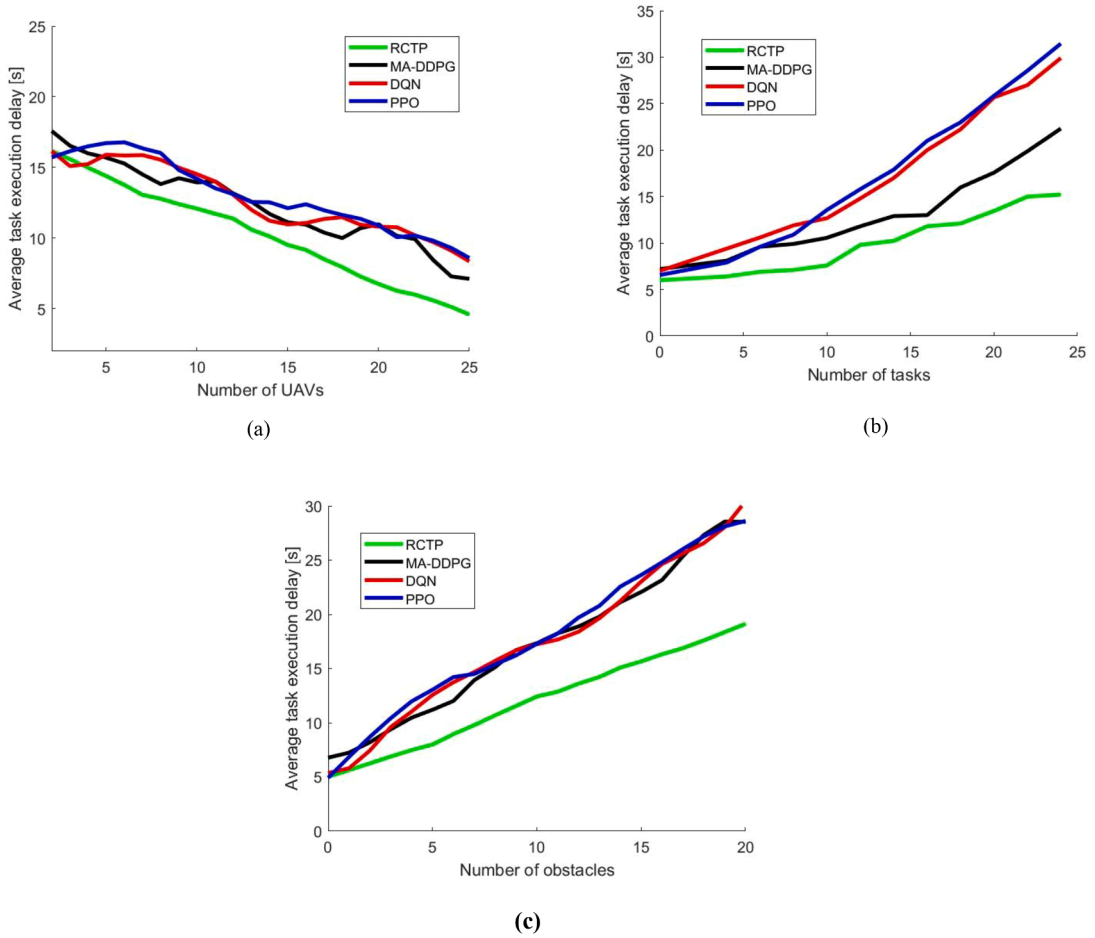


Fig. 9. Average task execution delay: (a) varying the number of UAVs, (b) varying the number of tasks, and (c) varying the number of obstacles.

Importantly, the performance gap becomes **larger** as the number of UAVs, tasks, and obstacles increases, indicating strong scalability and robustness under complex disaster-response conditions. Faster convergence and higher cumulative rewards highlight the benefit of MAML-based meta-initialization**,** while higher task completion rates and lower execution delays confirm the effectiveness of resource-aware coalition formation and priority-driven task scheduling**.** RCTP's meta-learning design enables rapid adaptation to previously unseen disaster scenarios represented as new combinations of task types, obstacle layouts, and communication conditions encountered during meta-training. However, as with prior meta-learning approaches, performance may degrade under strongly OOD scenarios involving fundamentally different dynamics. Importantly, RCTP is designed for rapid near-distribution adaptation, which is most relevant in real disaster response, where new environments typically share structural similarities with past experiences. This trade-off is explicitly acknowledged, and future work will focus on extending the meta-training distribution and incorporating task-robust meta-learning variants to further improve OOD generalization.

Additional results indicate that resource-aware load balancing and continuous-control path planning enable RCTP to maintain stable energy efficiency, fairness, and safety as task density and obstacle complexity increase. Compared with baseline methods, RCTP exhibits smaller performance degradation as system complexity grows, indicating more effective coordination and collision avoidance. RCTP follows a centralized training-decentralized execution paradigm, where a centralized critic is used only during training and UAVs operate independently at runtime. This design significantly reduces communication overhead, avoids single points of failure, and supports scalable operation under intermittent connectivity, making it well suited for large UAV swarms in disaster environments. The framework is also designed for lightweight online execution**,** relying on shallow network inference and local neighbor interactions, which supports real-time deployment on resource-constrained UAV platforms. The simulations further highlight practical challenges in large-scale disaster response, including intermittent LoS/NLoS communication, GPS degradation, tight timing constraints in dense environments, and sensitivity to unmodeled factors such as wind, smoke, and sensing noise. Although environmental effects such as severe weather and turbulence are not explicitly modelled, their impact is indirectly reflected through increased path cost, sensing uncertainty, and communication degradation. Explicit modelling of weather-aware dynamics and energy penalties is therefore identified as an important direction for future work. To validate the stability, we perform a one-at-a-time sensitivity analysis on the coalition suitability-score weights $(w_1, \ldots, w_4)$ and the task-allocation coefficients (α, β, λ). Each parameter is varied independently

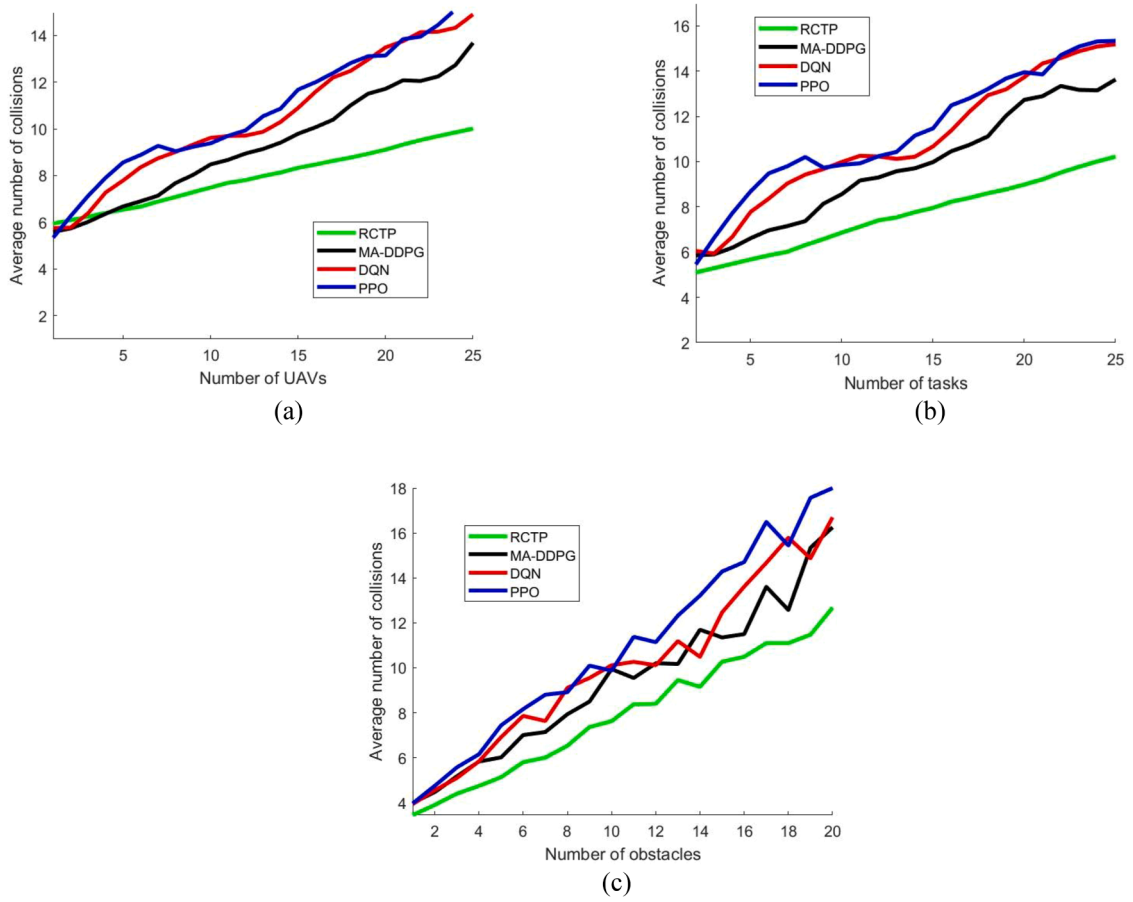


Fig. 11. Average number of collisions: (a) varying the number of UAVs, (b) varying the number of tasks, and (c) varying the number of obstacles.

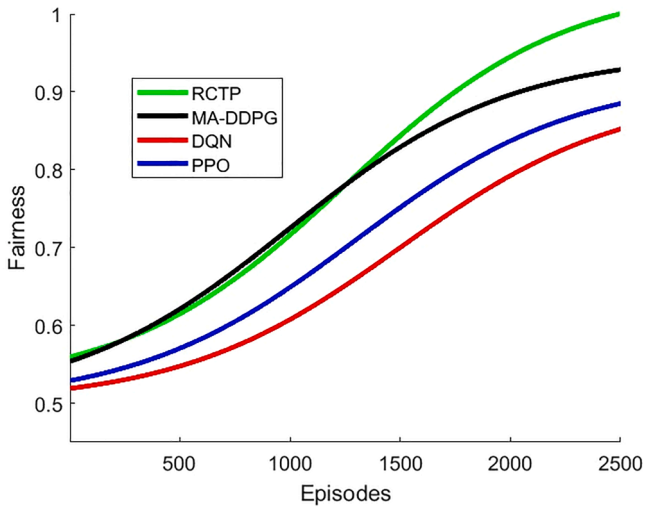


Fig. 10. Fairness versus episodes.

within ±30% of its nominal value while keeping all other parameters fixed using the same simulation settings. Each configuration is evaluated eight times using independent runs. Across all tested variations, the task success rate changes by less than 3%, confirming that RCTP remains stable under moderate parameter changes and does not rely on precise parameter tuning.

Finally, although more cooperative MARL methods exist, we compare RCTP with DQN, PPO, and MA-DDPG because these baselines

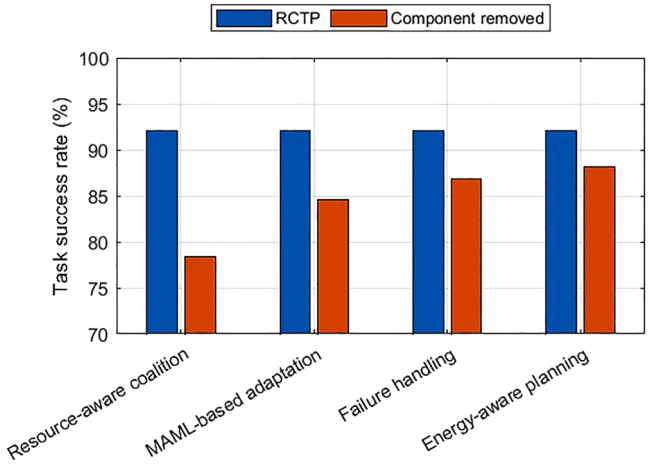


Fig. 12. Ablation study of key RCTP components.

are widely used in UAV coordination and path planning and represent the main RL categories of value-based learning (DQN), on-policy optimization (PPO), and CTDE multi-agent continuous control (MA-DDPG). While MAPPO [42] is a strong baseline, it is a multi-agent extension of PPO; therefore, including PPO provides a meaningful on-policy reference in our evaluation. We adopt MA-DDPG as the primary multi-agent baseline because RCTP is built on an MA-DDPG-style CTDE continuous-control backbone to jointly support coalition formation, task assignment, and UAV path planning under dynamic constraints. Moreover, MA-DDPG's centralized training captures inter-agent coupling while decentralized execution supports real-time operation on resource-limited UAV platforms, which aligns with the training and deployment setting targeted by RCTP. Therefore, MA-DDPG is the most suitable baseline to compare with our study. It allows us to clearly measure the contribution of our modules (MAML-based meta-learning, resource-aware coalition formation, priority-aware task assignment and execution, path planning, and failure handling). It also keeps the evaluation practical because it has manageable critic complexity, scales better, and supports real-time execution on resource-limited UAV platforms. Overall, these findings position RCTP as a practical, scalable, and adaptive framework for real-time multi-UAV disaster response, while clearly delineating current limitations and promising avenues for future research, including richer environmental modelling, broader meta-training for improved OOD robustness, larger swarm evaluations, and real-world field deployment.

Conclusion and future works

This paper presented RCTP, a resource-aware, adaptive coordination framework for multi-UAV disaster response that integrates MAML with MADDPG to achieve rapid policy adaptation and stable continuous-action control. By jointly addressing coalition formation, priority-aware task assignment, and energy-efficient trajectory planning within a unified learning framework, RCTP enables UAV swarms to operate robustly under dynamic tasks, limited onboard resources, intermittent communication, and platform failures. Simulation results across varying swarm sizes, task loads, and obstacle densities demonstrate consistent reductions in task completion time and energy consumption, along with improved robustness and safety compared to strong reinforcement-learning baselines, highlighting the framework's practicality for time-critical and resource-constrained scenarios.

Our future work will extend RCTP to more diverse and challenging disaster scenarios, with a strong focus on improving OOD generalization through stronger meta-learning strategies. This includes enabling faster and more reliable adaptation to disaster conditions that differ significantly from those seen during training. We also plan to scale the framework to larger and more heterogeneous UAV swarms, including multi-agency operations. In addition, environmental factors such as wind, severe weather, and reduced visibility will be explicitly modeled to enhance realism. Finally, we will conduct real-world field experiments and scenario-based case studies to validate scalability, communication robustness, and practical deployment feasibility. A direct experimental comparison with MAPPO is also identified as an important future extension.

CRediT authorship contribution statement

Sabitri Poudel: Writing – original draft, Visualization, Software, Resources, Investigation, Formal analysis, Data curation. **Sangman Moh:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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